

Chapter 5: Differentiation

Section: Mean Value Theorems

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

Local Extrema and the Derivative

The Mean Value Theorems

Monotonicity from the Sign of the Derivative

Summary

Motivation

From Local to Global

- $f'(x)$ captures the **local** behavior of f near a single point x

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- Question: if we know f' at *every* point of an interval, what can we say about f **globally**?
- Answer: **mean value theorems** — the bridge from local to global
- This section: Theorems 5.8 (Fermat), 5.9 (generalized MVT), 5.10 (MVT), 5.11 (monotonicity)



Local Extrema and the Derivative

Definition 5.7: Local Maximum and Minimum

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Let f be a real function defined on a metric space X . We say that f has a **local maximum** at a point $p \in X$ if there exists $\delta > 0$ such that

$$f(q) \leq f(p) \quad \text{for all } q \in X \text{ with } d(p, q) < \delta.$$

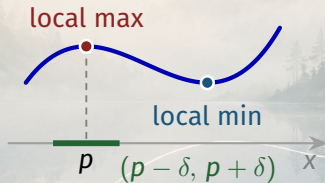
Local minima are defined likewise (with $f(q) \geq f(p)$).

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Theorem 5.8: The Derivative Vanishes at Interior Extrema

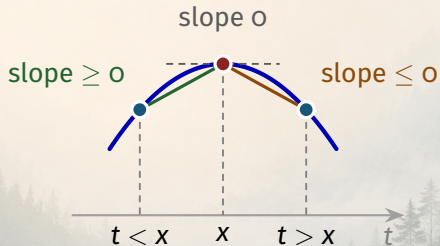
Theorem 5.8

Let f be defined on $[a, b]$. If f has a local maximum at a point $x \in (a, b)$, and if $f'(x)$ exists, then

$$f'(x) = 0.$$

The analogous statement for local *minima* is also true.

Theorem 5.8: The Derivative Vanishes at Interior Extrema



Key idea: compare the **one-sided** difference quotients at the maximum.

Proof of Theorem 5.8

Step 1: Set up the neighborhood

Choose δ as in Definition 5.7 so that $a < x - \delta < x < x + \delta < b$; on this neighborhood, $f(t) \leq f(x)$.

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Step 2: Approach from the left $\Rightarrow f'(x) \geq 0$

If $x - \delta < t < x$, then $\frac{f(t) - f(x)}{t - x} \geq 0$. Letting $t \rightarrow x$ gives $f'(x) \geq 0$.

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If $x < t < x + \delta$, then $\frac{f(t) - f(x)}{t - x} \leq 0$, so $f'(x) \leq 0$.

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If $x < t < x + \delta$, then $\frac{f(t) - f(x)}{t - x} \leq 0$, so $f'(x) \leq 0$.

Hence $f'(x) = 0$. ■



The Mean Value Theorems

Theorem 5.9: Generalized Mean Value Theorem

Theorem 5.9 (Generalized MVT)

If f and g are continuous real functions on $[a, b]$ which are differentiable in (a, b) , then there is a point $x \in (a, b)$ at which

$$[f(b) - f(a)] g'(x) = [g(b) - g(a)] f'(x).$$

- Differentiability is **not** required at the endpoints a, b
- Also known as the *Cauchy mean value theorem*

Proof Strategy: Theorem 5.9

Goal: Find $x \in (a, b)$ where a certain derivative vanishes.

Approach:

1. Combine f and g into a single auxiliary function h
2. Arrange $h(a) = h(b)$ — the increments cancel by design
3. h attains a maximum and a minimum on $[a, b]$ (Theorem 4.16)
4. If h is not constant, some extremum is *interior*; apply Theorem 5.8

Recall: Theorem 4.16

A continuous real function on a compact set attains its maximum and minimum.

Proof of Theorem 5.9

Step 1: The auxiliary function

Put

$$h(t) = [f(b) - f(a)]g(t) - [g(b) - g(a)]f(t) \quad (a \leq t \leq b).$$

Then h is continuous on $[a, b]$ and differentiable in (a, b) .

Proof of Theorem 5.9

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Then h is continuous on $[a, b]$ and differentiable in (a, b) .

Step 2: Equal endpoint values

$$h(a) = f(b)g(a) - f(a)g(b) = h(b) \quad (12)$$

It remains to show: $h'(x) = 0$ for some $x \in (a, b)$.

Proof of Theorem 5.9

Step 2 (recap)

$h(a) = h(b)$ by (12); goal: $h'(x) = 0$ for some $x \in (a, b)$.

Step 3: Case analysis

- If h is *constant*: $h'(x) = 0$ for every $x \in (a, b)$. ✓
- If $h(t) > h(a)$ for some $t \in (a, b)$: h attains its **maximum** at some x (Theorem 4.16); by (12), $x \in (a, b)$, so Theorem 5.8 gives $h'(x) = 0$. ✓

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- If $h(t) > h(a)$ for some $t \in (a, b)$: h attains its **maximum** at some x (Theorem 4.16); by (12), $x \in (a, b)$, so Theorem 5.8 gives $h'(x) = 0$. ✓
- If $h(t) < h(a)$ for some $t \in (a, b)$: same argument with the **minimum** of h . ✓

$h'(x) = 0$ unpacks to $[f(b) - f(a)] g'(x) = [g(b) - g(a)] f'(x)$. ■

Theorem 5.10: The Mean Value Theorem

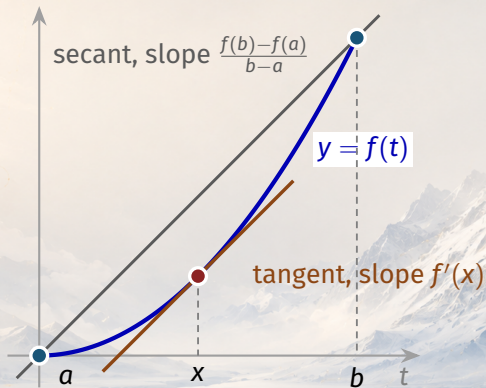
Theorem 5.10 (Mean Value Theorem)

If f is a real continuous function on $[a, b]$ which is differentiable in (a, b) , then there is a point $x \in (a, b)$ at which

$$f(b) - f(a) = (b - a)f'(x).$$

Some tangent line is **parallel to the secant line** through the endpoints.

Theorem 5.10: The Mean Value Theorem



Proof of Theorem 5.10

Proof

Take $g(t) = t$ in Theorem 5.9. Then $g(b) - g(a) = b - a$ and $g'(x) = 1$, so

$$[f(b) - f(a)] \cdot 1 = (b - a)f'(x). \quad \blacksquare$$

Technique: prove the general two-function statement first; the classical result is the special case $g(t) = t$.



Monotonicity from the Sign of the Derivative

Theorem 5.11: The Sign of f' Controls Monotonicity

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Suppose f is differentiable in (a, b) .

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- (a) If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is **monotonically increasing**.
- (b) If $f'(x) = 0$ for all $x \in (a, b)$, then f is **constant**.

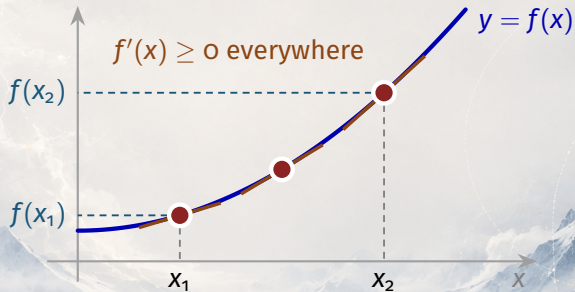
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- (a) If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is **monotonically increasing**.
- (b) If $f'(x) = 0$ for all $x \in (a, b)$, then f is **constant**.
- (c) If $f'(x) \leq 0$ for all $x \in (a, b)$, then f is **monotonically decreasing**.

Picture: Nonnegative Slope Means Never Decreasing



Proof of Theorem 5.11

One equation proves all three parts

Fix $x_1 < x_2$ in (a, b) . Apply Theorem 5.10 to $[x_1, x_2]$:

$$f(x_2) - f(x_1) = (x_2 - x_1)f'(x) \quad \text{for some } x \in (x_1, x_2).$$

Proof of Theorem 5.11

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Fix $x_1 < x_2$ in (a, b) . Apply Theorem 5.10 to $[x_1, x_2]$:

$$f(x_2) - f(x_1) = (x_2 - x_1)f'(x) \quad \text{for some } x \in (x_1, x_2).$$

Since $x_2 - x_1 > 0$, the sign of $f(x_2) - f(x_1)$ is the sign of $f'(x)$:

- (a) $f' \geq 0$ on $(a, b) \implies f(x_2) \geq f(x_1)$: **increasing**
- (b) $f' = 0$ on $(a, b) \implies f(x_2) = f(x_1)$: **constant**
- (c) $f' \leq 0$ on $(a, b) \implies f(x_2) \leq f(x_1)$: **decreasing**

The background is a soft-focus landscape featuring a calm lake reflecting distant mountains and cherry blossom trees. A complex, golden geometric pattern, resembling a string of pearls or a stylized constellation, is overlaid on the scene. This pattern includes various lines, dots, and circular motifs that create a sense of harmony and design. The overall color palette is warm and pastel, with pinks, blues, and golds.

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Key takeaways from this section:

- **Theorem 5.8 (Fermat):** at an interior local extremum where f' exists, $f'(x) = 0$

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 $[f(b) - f(a)] g'(x) = [g(b) - g(a)] f'(x)$, via an auxiliary function
- **Theorem 5.10 (MVT):** $f(b) - f(a) = (b - a) f'(x)$ — the bridge from local to global
- **Theorem 5.11:** sign of $f' \Rightarrow$ increasing / constant / decreasing

Coming up next: *The Continuity of Derivatives* — derivatives satisfy the intermediate value property, even when discontinuous!